

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (CL)

May 2012

CL306 Transportation Engineering-II

Date: 07.05.2012, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Give Classification of signals. [03]

(b) What is MAGLEV and tube railway? [04]

Q - 2 (a) Write functions of following railway track fittings with their neat sketches: [06]

(i) Dog spikes

(ii) Fishplates

(iii) Chair

(b) Calculate [08]

(i) Super elevation

(ii) Maximum permissible speed

(iii) Transition length

For a 3° curve on a high speed B.G. section with a maximum sanctioned speed of 110 kmph. Equilibrium speed is 80 kmph. Speed of the goods trains to be 50 kmph.

OR

Q - 2 (a) What are the types of gauges? Discuss about need of uniform gauges. [07]

(b) Discuss about various types of gradients used in track design. [07]

Q - 3 (a) What is points and switches? Explain left hand turn out with neat sketch. [07]

(b) Answer the following:

(i) Give necessity for geometric design. [03]

(ii) Draw neat and leveled sketch of permanent way. [04]

OR

Q - 3 (a) Write classification of railway stations and yards. [07]

(b) Explain scissors crossovers and diamond crossing with neat sketch. [07]

SECTION – II

- Q - 4 (a) Define the following terms: (Any Three) [03]
Littoral Drift, Dolphins, Moorings, Currents
- (b) What is meant by basic runway length? Discuss any one case to be considered with neat sketch. [04]
- Q - 5 (a) Write a short note on Minimum Turning Radius with neat sketch. [04]
- (b) How can the flying activities be classified? [04]
- (c) Define the following: (Any Six) [06]
Transit Sheds, Ware Houses, Apron, Runway, Taxiway, Payload, Aerodrome

OR

- Q - 5 (a) Define the following: (Any Four) [04]
Maximum gross take of weight, Apron, Zero fuel weight, Operating empty weight, Hanger
- (b) Write a short note on Airport Authority of India. [04]
- (c) Enlist and discuss any two factors which affect the necessary size of an airport. [06]
- Q - 6 (a) What is meant by breakwater? Enlist the types of breakwater and explain any one with neat sketch. [06]
- (b) Calculate the actual length of runway from the following data: [08]
Airport elevation : R.L. 100
Airport reference temperature : 28°C
Basic runway length : 650 m
Highest point along the length : R.L. 96.2
Lowest point along the length : R.L. 93.2

OR

- Q - 6 (a) Discuss any two types of harbour with neat sketch. [06]
- (b) Design an exit runway joining a runway and a parallel main taxiway. The total angle of turn is 35° and the maximum turn-off speed is 80 kmph. [08]
Assume the following: radius of entrance curve = 731 m, Runway width = 45 m, Taxiway width = 22.5 m
Suggest the maximum separation clearance and draw a neat sketch showing all the design elements
